

2025

INTERNSHIP REPORT



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PRIDE BANK

12/4/2025

Internship Experience

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Course: BSc Networking and Cybersecurity

Organisation: Pride Bank

Department: Business Technology

Internship Period: 20/10/2025 - 08/12/2025

Submitted to: Nicholas Aliganyira (Manager Networks and
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Approval

Supervisor's Name:

Position/Title:

Department:

Organization:

Supervisor's Signature:

Official Stamp:

Date:

EXECUTIVE SUMMARY

This report summarizes my seven-week internship at Pride Bank, where I rotated through key IT departments including Networking, Cybersecurity, Systems Administration, and IT Support., Core Banking etc. The internship aimed to give me practical exposure to enterprise IT operations and strengthen my technical and professional skills.

During the program, I participated in network troubleshooting, security monitoring, Linux and Windows system administration, and documentation of IT processes. I contributed to resolving technical issues, supporting system configurations, and assisting with internal security and operational tasks.

The experience enhanced my skills in networking, cybersecurity fundamentals, system administration, and professional communication. It also deepened my understanding of real-world IT environments and clarified my career direction. Overall, the internship was a valuable learning opportunity that significantly contributed to my growth as a future cybersecurity professional.

Chapter 1: Introduction

1.1 Background of the Organization

The Institution was founded in 1995 as Pride Microfinance Limited (PMFL) in 1995. A non-government organization with the support of the Norwegian Agency for Development Cooperation.

Its major objective was to offer credit to the poor, targeting those in the agricultural sector. In 1999, It was incorporated as a limited company and changed name to Pride Africa Limited. The Institution provided financial services to the Ugandan population who are unable to access financial services through Ugandan Commercial Banks. Its focus was the micro, small and medium entrepreneurs. In 2003, the Uganda government acquired 100 percent shareholding in the enterprise, changing the name to Pride Microfinance Limited Uganda. In 2005, it attained the status of a Tier 3 (an MDI) and on 29th April 2025, Pride Microfinance officially rebranded to Pride Bank Limited, a Tier 2 credit institution, following appropriate licensing by Bank of Uganda on 26 November 2024. Jachan Omach was introduced as the new chairman of Pride Bank and Edward Nkangi was introduced as the new executive director of the bank.

The bank currently has 47 branches across the country and an estimated core capital of around US\$50.0 million or UGX 174 billion.

1.2 Introduction to the business Technology Department

Business technology is the application of Information technology to improve business operations. The role of the business technology department is to enable the organisation achieve its strategic goals through use and management of the information technology resources to improve business operations and to boost overall productivity and efficiency of work.

It encompasses a wide range of digital tools, platforms and systems, from basic software to advanced solutions such as Manage Engine, Active Directory, Network Access Control etc.

All these systems are the integration of technology into every facet of the organisation to enhance efficiency, facilitate communication and to drive Innovation.

At Pride Bank, the key functions of the Business technology department include the following;

- IT operations
- Networks and Infrastructure
- Core Banking
- Alternate Banking Channels
- Innovations and Automations
- Information Security

1.3 Purpose and Objectives of the Internship

My main purpose for this internship was to learn and understand how information technology is handled in the banking sector, how the different systems work together for the organisation to operate effectively, and to observe how Information Security/ cybersecurity is handled in a banking environment.

Academically, the internship fulfills the academic requirement of relevant industry exposure before graduation and helps me to relate what I have been taught in the lecture room, to what is currently happening in the industry of Information Technology and Security.

Chapter 2: Internship Roles and Responsibilities

2.1 Position and Duration

My official internship title was Business Technology Intern and my internship officially started on the 20/10/2025 and ended on the 08/12/2025.



2.2 Key Responsibilities and Daily Activities.

At the bank on a daily basis, I had the responsibility of working with the network and infrastructure team.

The tasks I assisted in included; software installation on newly acquired machines, disposing off old machines, asset movement, trouble shooting and fixing issues with the network and infrastructure teams.

On a weekly basis I was required to engage with different teams and stakeholders within business technology for example; the Enterprise Systems Administrators, Network administrators, Core Banking Specialists, service Desk, Security Officers, Developers etc. To understand what they do, how they do it, and the importance of their work to the Business Technology Department and the organisation at large.

Below is a summary of the time I spent with the different teams within Business Technology during the course of the weeks of my internship.

Week 1: Infrastructure Department

On arrival, I was first introduced to the members of the Business Technology (BT) department as a new intern. Using an organogram, I was given a good visual description of the structure of the department and the relationships between the different roles and teams in BT.

During this week I was introduced to Information technology infrastructure management, which basically deals with how the hardware is handled from the time it is procured, how it is maintained during use in the organisation's environment and finally disposed off after use.

The tasks I carried out in this week included;

Installing baseline software needed for day-to-day operations on newly acquired machines.

Looked at repair request forms to vendors and how the repair process is handled for assets.

Disposed off old machines by erasing their hard drives, labelling them as disposed and placing them in storage where they will be picked up for discarding through processes like action.

Below are some photos:

Erasing hard disks



Replacing a hard disk in an old computer



An asset movement form.

PRIDE BANK LIMITED

SERIAL NO: 1745
DATE:

ASSET MOVEMENT REGISTER

Name of the Officer Requesting for Asset Movement: _____
Title: _____ Department / Branch: _____
Reasons for Asset movement: _____

No.	Quantity	Particulars of the Asset(s) To be moved:	Serial Number / Engraved number (Where applicable)
1.			
2.			
3.			
4.			

Destination of the Asset(s): _____ Expected Date of Return: _____
HEAD OF DEPARTMENT / SECTION / SUPERVISOR / BRANCH HEAD / DESIGNEE
Name of Authorizing Officer: _____ Title: _____
Signature: _____ Date: _____
Remarks: _____
Name of Releasing Officer: _____ Title: _____
Signature: _____ Date: _____
Remarks: _____

ASSET(S) TAKEN BY:

Branch / Department: _____
Name: _____ Title: _____
Signature: _____ Date: _____
Remarks: _____

ASKARI (SECURITY PERSONNEL)

Name: _____ Date: _____
Signature: _____

ASSET(S) RECEIVED BY:

Branch / Department: _____
Name: _____ Title: _____
Signature: _____ Date: _____
Remarks: _____

Week 2: System administration

In the second week, along side my daily infrastructure duties, I spent time with the Enterprise systems administrators.

During this week, I was introduced to the concept of active directory. The corner stone of modern Enterprise systems management. It is used for organising and managing users and devices based on different attributes like groups.

The core concept of AD include; Forests, Domains, Organisational Units and Groups.

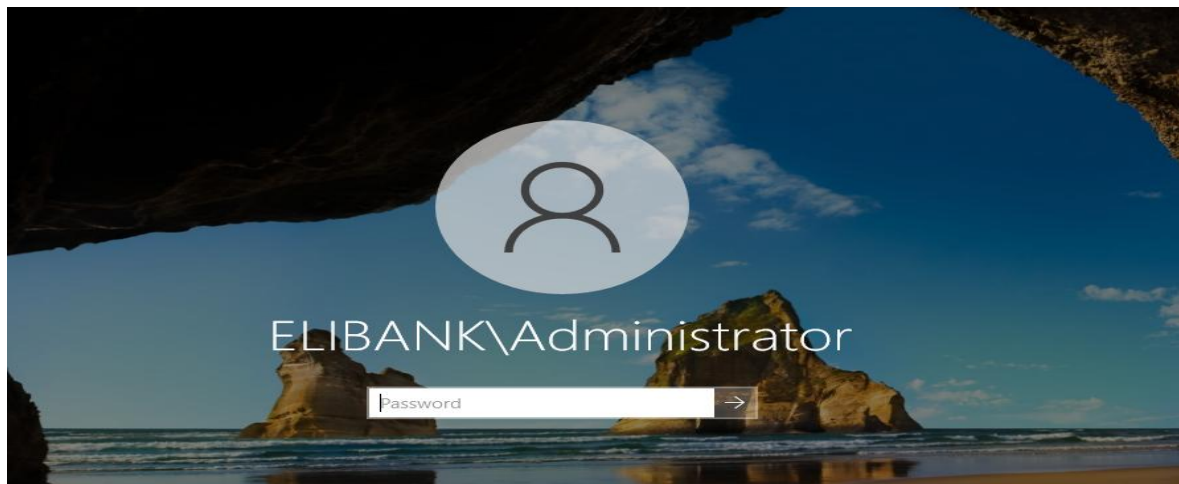
The 2 main functions of Active Directory.

- Authentication - used to verify users, computers and other resources on the network.
- Authorization - Used to authorize access to specific resources on the network. This is done through assigning permissions and rights to users and groups which determine what they are allowed to do on the network.

At the end of the week, I created an AD environment, added users and a device.

In this week, I also went to the server room and participated in taking the monthly backup.

ELIBANK the name of the domain for the Active directory I created.



Week 3: Networks Department.

In my third week, alongside my daily infrastructure duties, I spent time with the Network administrators. These are the individuals in the department that are responsible for managing enterprise network.

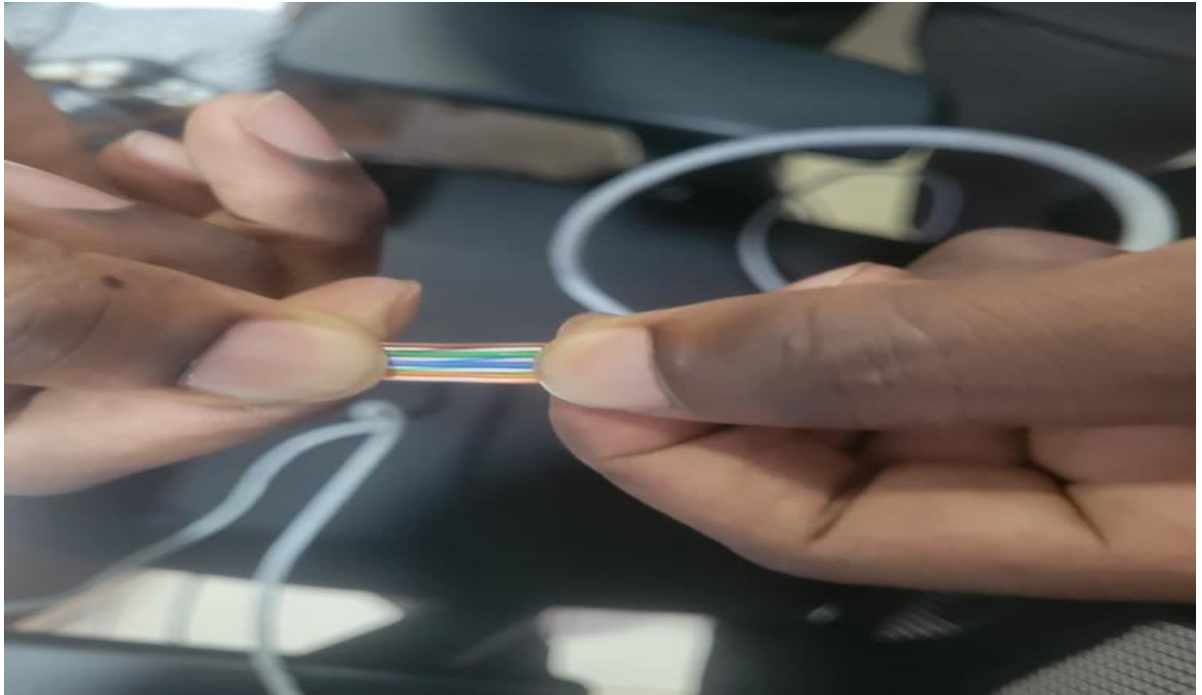
Their goal is to maintain availability of the network at all times. This ensures all business operations and communications take place smoothly without interruptions.

My tasks this week were to;

- Understand what makes up an enterprise network from the routers to switches to firewalls, etc.
- Understand Virtual local area networks and network segmentation (Vlans)
- Understand Access Control Lists (ACLs) and at which point they are appropriately applied. e.g. towards the source or towards the destination.
- Terminated and test network cables before they are added to the environment

Below are some pictures from this week

Used T-568B colour code and cat 6A cable and an RJ45 to make a network cable.



Testing the cable



RJ45

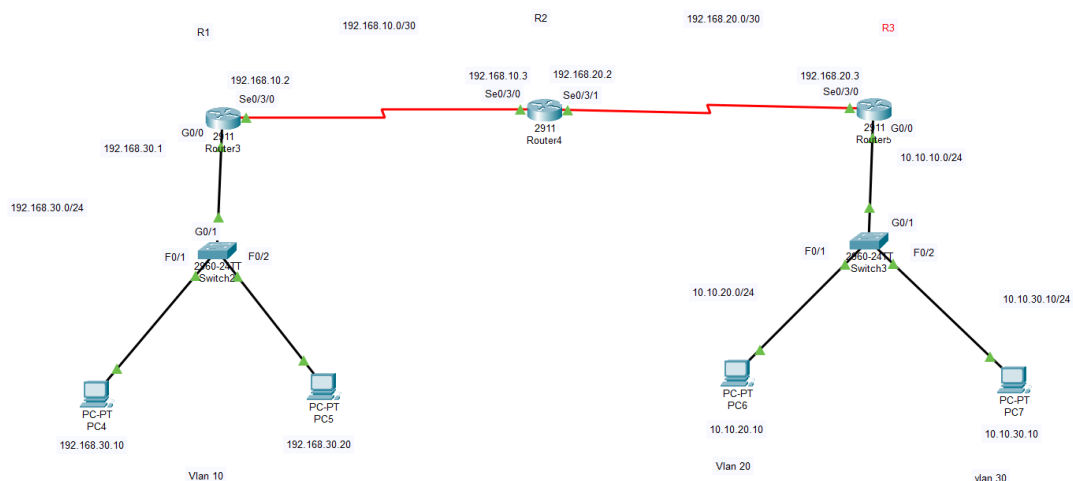


Week 4: Networking Department. Configurations of routers and switches.

In this week, I was given access networking hardware. This includes routers and switches. In the previous week I had learnt about Vlan's and ACLs so the task given to me this week was to configure Vlan's, ACLs and a routing protocol of my choice on the hardware given to me.

Below are some images from that week's activities.

The network diagram.



The hardware I used. 2 (2911 cisco routers), 2 (2960 cisco switches) and 1 (1941 cisco router)



Connected the console cables to the devices and my PC to enable me access them to carry out the configurations on my lap top



Configuration process.



Picture taken after successfully configuring the devices



Week 5: Innovation and Automation.

In this week, I spent my time with the Innovations and automations department. The role of this team in business technology is to drive business growth and efficiency by identifying, developing and implementing new ideas and technologies to streamline operations.

This department fosters the culture of continuous improvement across the organisation by developing various inhouse applications like a human resource management system (HRMS) and integrating new technologies into the business. For example; escrow with companies like MTN, integrating SWIFT for international transactions etc.

My personal goal during this week was to understand how applications in particular web applications are developed. The team did a great job of explaining to me what makes up a web application i.e.

- Client side / front end
- Server side / back end
- The databases
- Application Programming Interfaces (APIs)

All the above components come together to bring a web application to life. I was also introduced to the concept of web application frame works and software design patterns like MVC.

A web application is a software program that can be accessed through a web browser and does not require installation on a user's device. It's main advantage unlike a traditional website is that it allows users to send and process data with a server unlike the traditional ones which are static and primarily display information. Examples of web applications include online banking where you can login, perform actions and interact with data.

MVC stands for Model View Controller. It is a design pattern used to organise code by separating data logic, user interface display and the application's control flow.

My task this week was to understand the above concepts and come up with a simple web application with CRUD capabilities and to test it for a common web application vulnerability.

C - create

R - read

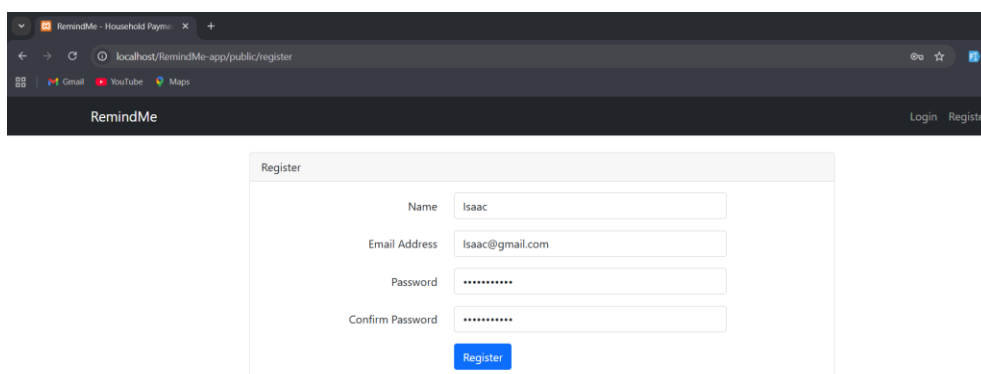
U - Update

D - Delete

Below are some pictures from that project.

I came up with an application for making simple reminders for payments. RemindMe

Creating a new ser



Register

Name Isaac

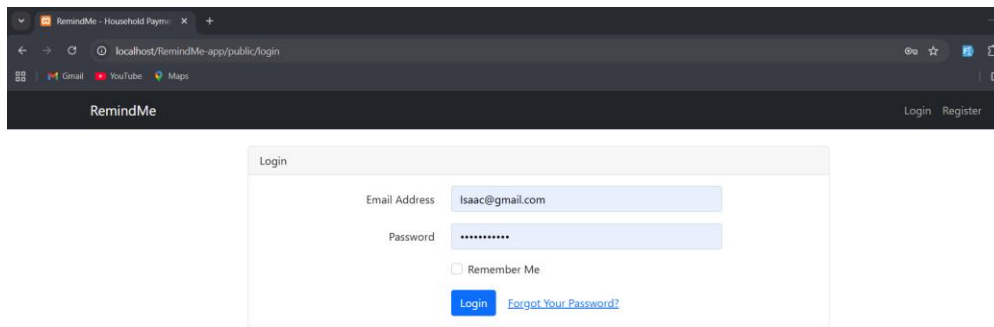
Email Address Isaac@gmail.com

Password

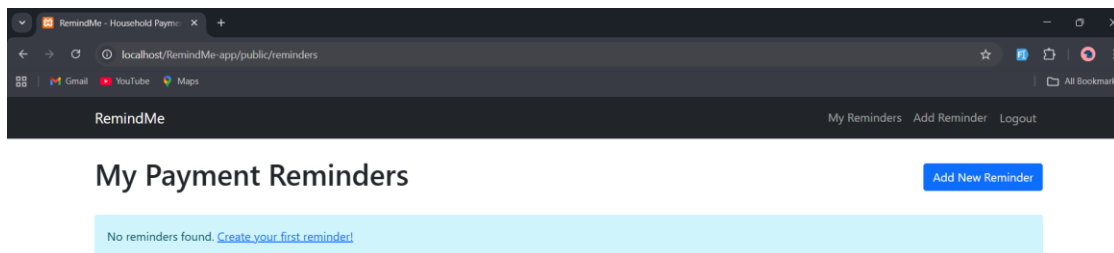
Confirm Password

Register

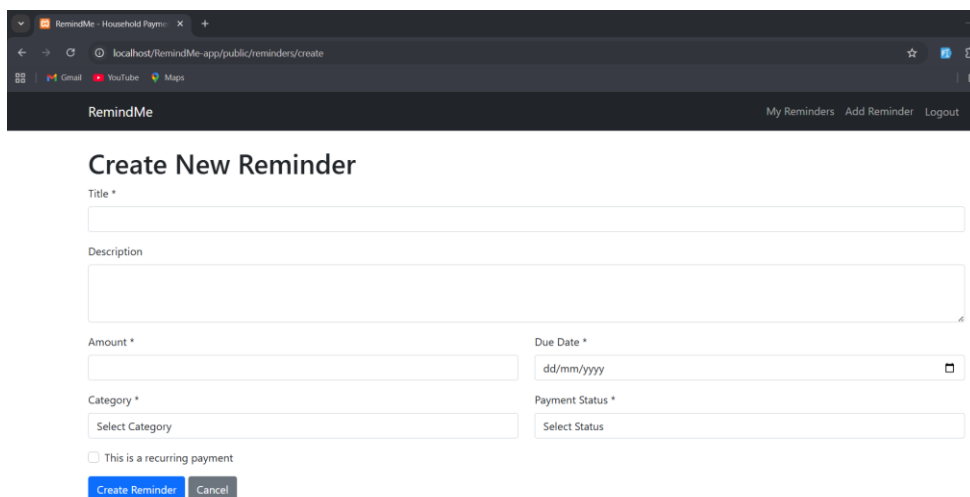
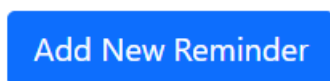
User login



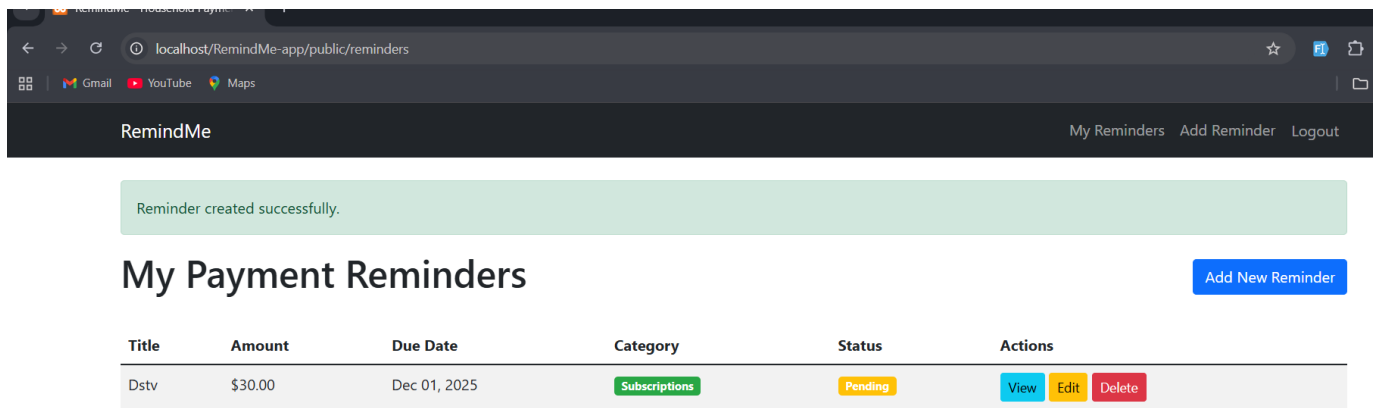
Simple login page.



Creating a new reminder



How a complete reminder looks. In a users account.



Week 6: Core Banking and Service Desk.

In this week, I interacted with the core banking and service desk team. My goal this week was to understand what core banking is and how the service desk team supports the different users in the organisation

Core banking. This is the centralized back-end system that handles the bank's daily operations such as processing transactions, managing accounts and processing loans. This is the system used by the bank at the branches to allow customers access their accounts at all branches. This possible because it centralizes data sharing. This is important because it maintains consistency in service delivery.

The core banking teams role is to serve as a bridge between banking operations and technology, using languages like java to update the legacy systems, guiding the bank through processes like migrating from legacy systems to modern platforms and providing continuous technical support to end-users the bank employees. This ensures that they are trained and familiar with the system and its features.

Service Desk. This is the point of contact for the users to get IT support. By the users I am referring to are the different employees of the bank form here at HQ to those at the various branches. Their key roles include but are not limited to; service request fulfilment, user support, assignment of tickets and communication.

I participated in the process of handling user tickets.

Week 7: Information Security/ Cybersecurity

Information Security/ cybersecurity refers to the protection of information and information systems from unauthorised access, use, disclosure, disruption, modification and destruction.

The main goal of this team is to ensure the Confidentiality, Integrity and Availability of data.

My goal for spending time with this team was to practically see and understand how what I have been learning in the class room is applied and managed in the bank's enterprise environment.

The first task I had this week was to read the BT Enterprise Technology, Security and governance policy. This is a detailed manual which establishes mandatory security, operational, and governance requirements for all staff, contractors, and affiliates accessing Pride's systems and data.

Major policy areas include rigorous standards for Computer Security, detailed procedures for Data Backup and off-site storage, and mandates for Cyber Security measures like annual penetration tests. Furthermore, the document establishes clear guidelines for the use of company resources, such as strict Email and Internet Policies, while also specifying the formal structure of a Change Control Committee and the activation protocols for the Disaster Recovery Plan (DRP). It sets forth specific roles and responsibilities for managing Information Ownership and ensuring stringent physical and digital security across all data centres and network infrastructure.

We went through concepts like network security and its various components like Network Access Control (NAC), Virtual Private Networks (VPNs) web application security, Identity and Access Management (IAM). Endpoint security, Security Information and Event Management (SIEM)

During this week, the highlight for me was participated in making of the daily security report which provides a detailed view of what happened the previous day in regards to Information security.

This report includes key feeds like; Anti-virus software monitoring reports gathered from the agents on the end points, Review of Active Directory, Review of Core Banking, Patch Management Update, Review of Alternate Banking Channel systems e.g. ATMs etc.

2.3 Major Projects Undertaken:

Active Directory

Project Name: Eli Bank AD

Objective: To Understand how Active Directory is set up and how Users and Devices are managed

Technologies: Oracle Virtual box, Windows Server, Windows 11 and 10, Active Directory.

Outcome: Developed a functioning active directory environment, added users and devices to it



AD Project.pdf

Network Configurations

Project Name: Network configurations

Objective: To successfully configure Vlans, a routing protocol and Access Control lists.

Technologies used; Cisco Packet Tracer, Cisco 2911 routers, 2960 switches and a 1941 router, console cables, cat 6A cable, a testing and clipping tool.

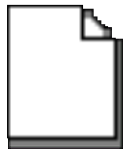
Outcome; Successfully reset default configurations on the above equipment, Configured Vlans, OSPF routing protocol and Access control Lists (ACLs) on the switches and routers.

Web Application Development with Laravel

Project Name: RemindMe App

Objective: To develop a simple functioning web application using the MVC development pattern, laravel web application frame work, with CRUD capabilities and to test it for a web application vulnerability.

Technologies used: VS code, Laravel herd, Xampp server, kail Linux and burpsite.



RemindMe-app Project.pdf

Outcome; Learnt how web applications are developed from the back end to the front end. Input sanitization against SQLi

Drafting of a Security status report

Project Name: Daily Security Report

Objective: To Understand what a security report contains, what kind of updates are made in the report and what items to look out for when reporting.

Technologies used: Kaspersky, AD 365 and AD info, Identity Service Engine (ISE), Ms excel, Ms word etc.

Outcome: Successfully helped in drafting the daily security report and gained understanding of the key areas to lookout for when maintaining the security of the bank's information systems.

Chapter 3: Analysis of Learning Experience.

3.1 Technical Skilled Acquired

Network configuration; Virtual Local Area Networks (Vlans), Access Control Lists (ACLs), Routing Protocols e.g. OSPF

Software and Tools: Putty for configuring networking devices, Visual Studio Code as a development environment (IDE), Xampp for local hosting, MySQL for data basement, Laravel for development in PHP, Virtual box for Virtualization, Windows Server for hosting the Domain Control and setting up Active Directory.

Banking Systems: Exposure to core banking systems like Rubicon, Management Systems like Manage Engine, Kaspersky for end-point security.

Methodologies: During my work I used both Agile and waterfall methods since I handled a number of tasks.

3.2 Professional Skills Developed

Communication: Through my numerous interactions with my team members and other stakeholders.

Teamwork and Collaboration: Working in a cross-functional team.

Problem-solving: Gained though various trouble shooting tasks as I worked with the team and on my personal projects.

Time Management and Prioritization: We juggled multiple tasks in a fast-paced environment.

Understanding of Corporate Culture: Gained understanding of how a large, regulated organization like a bank functions.

3.3 Integration of Academic Knowledge

During my internship period, the knowledge from my University Courses for example Networking and Information security Compliance and Governance helped me to easily understand the Business Technology Policies followed in the organisation like Password policies, Incident response, access management, backup procedures, disaster Recovery etc.

The knowledge on networking was crucial for understanding subnets and subnetting during IP addressing, the various networking protocols used like Https, SSH etc.

This helped me gain a firm understanding of the above concepts during my time at the bank.

Chapter 4: The Organisation and the Banking Technology Landscape

4.1 Analysis of the Department's Function.

The business technology department is a strategic partner that manages and innovates the IT infrastructure, systems and digital process to support business goals, ensure secure and reliable operations and drive digital transformation.

All the teams work together like a well-oiled machine to ensure communication is effort less across this large organisation and operations like core banking processes are carried out with out disruption.

The BT team is also always constantly interacting with other teams in the organisation like the finance department, marketing department, procurement department to mention but a few. This is in an effort to support them as they navigate the use of the various technologies used in the organisation and to also find ways of optimizing their work by developing in house applications to automate processes hence improving efficiency of work.

4.2 Key Industry Trends Observed:

Digital Transformation: There is a great shift to online and mobile banking. These are referred to as the alternate banking changes. The bank has made it easy to access their services with convenience without the hustles of dealing with banks the old school way like showing up physically to a branch with long queues or going to an ATM and having your debit card swallowed by the machine.

Data Analytics and AI: Data is constantly being gathered and analysed to get customer insight for developing new products and to detect fraud which is a big challenge to these financial institutions.

Cybersecurity: Alot of investment is being made into safe guarding the bank's Information Systems and technologies. This security is no longer a concern of the Business Technology Department alone but has been a responsibility to all the stake holders. Starting from other staff in the bank to even the customers the bank serves who are educated to safe card their bank details. Grooming a deep culture of information security hygiene in the organisation.

API Integrations: Different technologies have to communicate for the bank's different channels to communicate to foster use of various channels of operations. These APIs have to

be highly secured and constantly reviewed to enable smooth operations and integrations of various technologies in the business.

Cloud Adoption: Some systems have been migrated to the cloud to ensure efficient use and management of resources.

Chapter 5: Challenges and Recommendations.

5.1 Challenges faced:

Due to the demanding nature of the various Business Technology team member's duties, I found it difficult to with engage particular teams during my quest for information.

How I overcame it: I scheduled particular meeting periods with the various team members I wanted to interact with prior. This made it easy for them to make time for me which was very effective for my learning.

5.2 Recommendations for the Organisation.

I recommend that the department develops a clear internship schedule for the interns easily engage with the team members since time is properly scheduled for interaction. This will ensure that the engage with everyone and that the team members are not just ambushed by the interns without prior knowledge.

5.3 Recommendations for the University.

The official duration of time allocated for Internship should be rescheduled to an appropriate time where students do not have to go back and forth between the University and their places of work.

This reduces disruptions and improves concentration of the students at work hence improving the internship experience.

Chapter 6: Conclusion

6.1. Summary of the Experience:

Overall, I have had a great time at Pride Bank. I am also grateful for the continuous support and engagement from the Business Technology team members. They treated me like one of their own and shared their great minds and various experiences with me, I am true grateful for the opportunity I have had to spend time with them, in which I have learnt and unlearned a number of things which wouldn't have been possible had I stuck to the traditional classroom setting.

6.2 Achievement of Objectives:

I achieved my primary goal for this internship, which was to learn and understand how information technology is handled in the banking sector, how the different systems work together for the organisation to operate effectively, and observation Information Security/ cybersecurity in a banking environment.

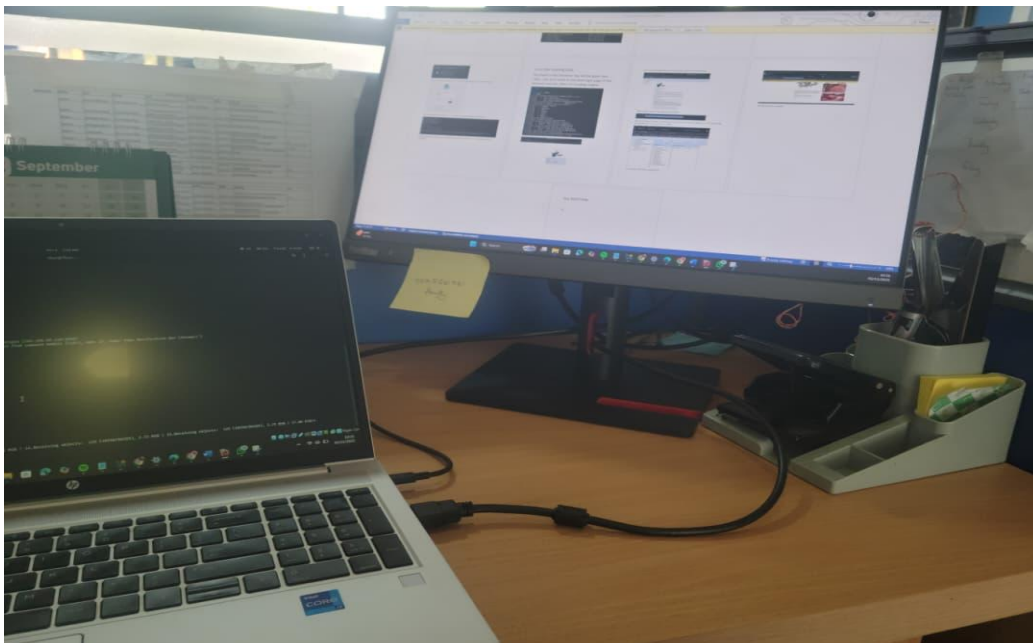
6.4 Future Plans:

I plan to use the experience and knowledge I have acquired from Pride bank to further prepare myself for my Information security career since I have seen the concepts taught in the lecture room in action in the organisation's environment.

6.5 Photos.



My work station.



Thank you, Pride Bank.



6.6 References;

<https://pridebank.co.ug/about.html>

https://en.wikipedia.org/wiki/Pride_Bank_Limited

